Fast Sampling for Joint Sky-Image and PV Diffusion Forecasting: A 15-Step DDIM Evaluation

Muhammad Umer
Department of Computer Science, SST
University of Management and Technology
Lahore 54770, Pakistan
f2023266912@umt.edu.pk

Haseeb Khan
Department of Computer Science, SST
University of Management and Technology
Lahore 54770, Pakistan
f20232661009@umt.edu.pk

Dr. Jameel Ahmad
Department of Computer Science, SST
University of Management and Technology
Lahore 54770, Pakistan
jameel.ahmad@umt.edu.pk

Dr. Abdul Hameed
Department of Computer Science, SST
University of Management and Technology
Lahore 54770, Pakistan
abdul_hameed@umt.edu.pk

***Abstract*—PV-MM-Diffusion, an end-to-end multimodal diffusion model that jointly generates future sky images and photovoltaic (PV) power, has been reported to improve 15-minute-ahead probabilistic PV forecasting with 60-step DPM-Solver++ sampling, a multi-GPU training budget and a test set of multi-cloud days. We study this model class under cheaper and more demanding conditions: a compact re-implementation that keeps the coupled U-Net, Random-Shift PV Multi-Modal Attention (RS-PV-MMA) and empty-frame placeholder training, is trained on a single 8 GB RTX 3070 and is sampled with a 15-step Denoising Diffusion Implicit Model (DDIM) solver. The model and all baselines are scored on the same 1,340 all-sky test windows of the SKIPP'D benchmark. A 30-member ensemble attains a continuous ranked probability score (CRPS) of $1.222\,\mathrm{kW}$, a 7.5% skill over smart persistence and an 18.4% reduction relative to a two-stage video-then-PV pipeline, at $155\,\mathrm{ms}$ per member; without sky images the CRPS rises to $2.072\,\mathrm{kW}$. However, the ensemble median has a higher mean absolute error than persistence (1.684 vs. $1.321\,\mathrm{kW}$) and a negative bias, and nominal 90% intervals cover only 77.9% of observations. Diffusion models therefore add value mainly through spread rather than sharper point forecasts, and calibration and bias are the main targets for future work.**

***Index Terms*—solar forecasting, photovoltaic power, sky images, diffusion models, DDIM, probabilistic forecasting, multimodal learning, CRPS**

I. Introduction

The output of a PV plant can drop by more than half within a minute when a cloud crosses the sun. Such ramps complicate grid balancing, reserve scheduling and battery dispatch, and they are poorly anticipated by numerical weather prediction or satellite forecasts, whose resolution is too coarse for intra-hour horizons [6]. Ground-based sky cameras observe the clouds that will shade the array in the next few minutes and have become the standard input for PV nowcasting [8], [11]. Most sky-image forecasters are deterministic regressors trained with a squared-error loss [8], [10]; under uncertainty about where a cloud edge will be, they regress to the conditional mean and give no measure of confidence, whereas grid operators need the distribution of future output [7].

Denoising diffusion models [13] draw such distributions by sampling plausible futures. Huang *et al.* [1] proposed PV-MM-Diffusion, which denoises future sky frames and future PV power jointly in one reverse process, and reported CRPS and Winkler-score reductions of 18% and 40% relative to a SkyGPT→U-Net pipeline. That evaluation, however, uses five multi-cloud test days with clear-sky samples removed, 60-step DPM-Solver++ sampling [25] and eight GPUs, and it reports probabilistic scores only. We therefore (i) re-implement PV-MM-Diffusion compactly, sample it with 15-step DDIM [14] and train it on one 8 GB GPU, documenting every deviation (Table I); (ii) evaluate it and all baselines on the same 1,340 all-sky SKIPP'D [2] test windows with one split and one set of metrics; and (iii) report both its best-in-study CRPS and its weaker point accuracy, calibration and image quality.

II. Related Work

Sky-image PV forecasting. Deep learning has replaced hand-engineered cloud-motion pipelines [11]: SUNSET [8] regresses 15-minute-ahead PV power from stacked sky images and PV history, and Paletta *et al.* [10] showed that such regressors often learn a smoothed persistence and struggle with ramps. Recent models fuse sky images with PV and meteorological series through physics-derived image features, a Swin-Transformer encoder and a temporal fusion transformer [4], or embed physical knowledge through physics-aware retrieval with a frozen foundation-model prior and distribution-shift correction [3].

Probabilistic and generative forecasting. Probabilistic PV forecasts are commonly obtained by quantile regression or parametric outputs [7] and verified with proper scoring rules [26]. Generative models instead sample future skies:

TABLE I
REFERENCE PV-MM-DIFFUSION STUDY [1] VS. THIS STUDY

Aspect	Reference [1]	This study
Test set	5 multi-cloud days, clear sky removed	20 all-sky days, 1,340 windows
Sampler	DPM-Solver++, 60 steps	DDIM (η=0), 15 steps
Backbone	2 blocks/scale, image self-attn.	1 block/scale, none
RS-PV-MMA	Image patches $\times$ PV tokens	Frame-pooled tokens
Training	8$\times$ RTX 3090, 10^5 steps	1$\times$ RTX 3070, early stop
Two-stage	SkyGPT$\rightarrow$U-Net	Light 3-D CNN$\rightarrow$CNN
VGG-CS	Best of 30 samples	First sample
CRPS / WS_{90} (kW)	2.63 / 21.46	1.222 / 10.50
Gain vs. two-stage	18%	**18.4%**

SkyGPT [9] generates stochastic sky videos with a physics-constrained VideoGPT and maps each to PV power with a separate U-Net, and SkyEvo [5] likewise derives probabilistic PV forecasts from generated sky-image evolution. PV-MM-Diffusion [1] removes the second stage by adapting the coupled U-Net and random-shift cross-modal attention of MM-Diffusion [16] to sky-image and PV sequences.

Diffusion models and samplers. DDPM [13] learns to reverse a Gaussian noising process and underlies diffusion models for video [15], probabilistic time series [17], [18] and precipitation nowcasting [19], [20]. A diffusion forecaster is sampled many times per issue time, so the sampler dominates its cost. DDIM [14] reuses the trained network with a non-Markovian sampler of tens of steps; the higher-order DPM-Solver++ [25], used by PV-MM-Diffusion, reduces the step count further. DDIM is the simplest option and requires no change to the trained network.

III. METHODOLOGY

The design follows PV-MM-Diffusion [1]; our departures are collected in Table I.

A. Problem Formulation and Training Objective

Let $\mathbf{I}_t \in \mathbb{R}^{64\times64\times3}$ be the sky image and $p_t \in [0, P_{\max}]$ the PV power at time t ($P_{\max} = 30$ kW), sampled every $\Delta = 2$ min. Given the history $\mathcal{H} = \{\mathbf{I}_{t-k\Delta}, p_{t-k\Delta}\}_{k=0}^{7}$, the model draws M samples of the future $\mathbf{x}_0 = \{\mathbf{I}_{t+k\Delta}, p_{t+k\Delta}\}_{k=1}^{8}$ (lead times 2–16 min) from $p_\theta(\mathbf{x}_0 \mid \mathcal{H})$; the PV components form the ensemble forecast. PV power and pixels are scaled to $[-1, 1]$. Following DDPM [13], the future block is corrupted as $\mathbf{x}_\tau = \sqrt{\bar\alpha_\tau}\,\mathbf{x}_0 + \sqrt{1-\bar\alpha_\tau}\,\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, with a linear schedule $\beta_\tau \in [10^{-4}, 0.02]$ over $T = 1000$ steps. The clean history is concatenated with the noisy future, and ϵ_θ predicts the noise of both modalities with the loss

$$\mathcal{L} = \mathbb{E}_{\tau,\boldsymbol{\epsilon}}\Big[\, m_p \left\|\boldsymbol{\epsilon}^p - \epsilon_\theta^p\right\|_2^2 + m_I \left\|\boldsymbol{\epsilon}^I - \epsilon_\theta^I\right\|_2^2\Big], \qquad (1)$$

where $m_p, m_I \in \{0, 1\}$ are loss masks. Since cameras and PV loggers fail independently, each training window is assigned with equal probability to one of three modes [1]: (i) ***SI+PV***, the full history; (ii) ***SI***, past PV replaced by zero placeholders; (iii) ***PV***, past images replaced by zeros and $m_I = 0$. Mode (i) gives the main results and mode (iii) the modality ablation.

B. Coupled U-Net Denoiser

Fig. 1 outlines the denoiser: two parallel four-level U-Nets [21] with widths $(16, 32, 64, 128)$ operate on the 16-frame sequence. Image blocks apply a 3×3 spatial convolution per frame, a kernel-3 temporal convolution per pixel and a residual shortcut; PV blocks apply two kernel-3 1-D convolutions and multi-head temporal self-attention [22]. A sinusoidal diffusion-step embedding is added in every block. To fit an 8 GB budget, each scale has one block per branch instead of two, and the image self-attention of the reference is omitted.

RS-PV-MMA. At levels 2–4 the branches exchange information through Random-Shift PV Multi-Modal Attention. The reference lets each PV token attend to the image patches of a randomly offset window of frames. We keep the local window and random offset but pool image features over space to one token per frame, then compute four-head cross-attentions PV$\rightarrow$image and image$\rightarrow$PV restricted to $|i - j| \le w$, with $w = 1, 4, 8$ at levels 2, 3, 4. Outputs are added residually (the image output broadcast over space). During training the image tokens are rolled in time by a random shift $s \in \{-2, \ldots, 2\}$. Pooling makes the cost independent of image resolution but prevents attending to the region around the sun (Section VI).

C. DDIM Sampling

At inference the future block starts from Gaussian noise and is denoised with DDIM [14] on an evenly spaced subsequence $\tau_S > \cdots > \tau_0 = 0$ of the training steps. With $\hat{\mathbf{x}}_0 = (\mathbf{x}_{\tau_i} - \sqrt{1-\bar\alpha_{\tau_i}}\,\epsilon_\theta)/\sqrt{\bar\alpha_{\tau_i}}$ and $\eta = 0$, each update is deterministic,

$$\mathbf{x}_{\tau_{i-1}} = \sqrt{\bar\alpha_{\tau_{i-1}}}\,\hat{\mathbf{x}}_0 + \sqrt{1-\bar\alpha_{\tau_{i-1}}}\,\epsilon_\theta(\mathbf{x}_{\tau_i}, \tau_i), \qquad (2)$$

so ensemble members differ only through their initial noise. We use $S = 15$ steps and clip sampled PV to $[0, P_{\max}]$. DDIM with $\eta = 0$ is the first-order solver of the probability-flow ODE that DPM-Solver++ integrates at higher order; as other factors also differ (Table I), we do not isolate the solver's effect.

IV. EXPERIMENTAL SETUP

Data and split. SKIPP'D [2] pairs 64×64 sky images at 1-minute resolution with the power of a 30 kW array at Stanford University (349,372 training and 14,003 test records). Windows of 16 frames at a 2-minute cadence are taken within a day, skipping logging gaps. A seeded 10% of *calendar days* is held out for validation to avoid leakage between overlapping windows. The test set has 1,340 windows over 20 clear, partly cloudy and overcast days, 1,115 of them daytime (mean output above 15% of capacity). The reference instead tests on five multi-cloud days of the SkyGPT partition [9], so only relative improvements are comparable.

Baselines. Smart persistence (SPM) holds the clear-sky index $k = p_t/p_t^{\mathrm{clr}}$ constant, $\hat p_{t+h} = k\,p_{t+h}^{\mathrm{clr}}$ [8], with p^{clr}

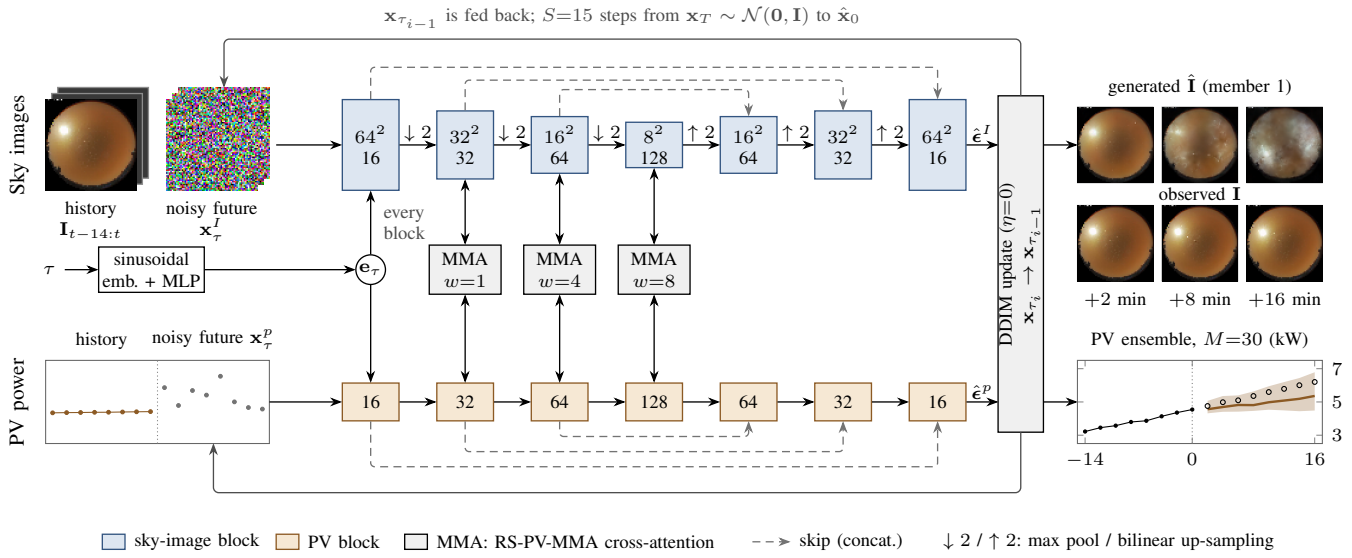


Fig. 1. Coupled U-Net denoiser and DDIM sampling (compact form of PV-MM-Diffusion [1]) on a real test window (2019-07-14 08:12). Inputs: 8 history frames and PV readings followed by 8 future frames that start as noise (PV in normalized scale). Image blocks are labelled with spatial size and channels, PV blocks with channels; RS-PV-MMA with temporal window w couples the branches at levels 2–4, and $\mathbf{e}_\tau$ enters every block. Outputs after 15 DDIM steps: generated frames of the first member vs. observed ones, and the 30-member PV forecast (band: 5–95%; line: median; dots/circles: observed history/future).

the Ineichen–Perez clear-sky irradiance from pvlib [12] scaled to 30 kW. The *two-stage pipeline* stands in for SkyGPT→U-Net: a 3-D CNN video predictor followed by a CNN that regresses PV from future frames and past PV, each trained with a squared-error loss and fed Stage 1's predictions at test time. The *PV-only ablation* samples the trained joint model in mode (iii) with all past images replaced by placeholders.

Implementation. Models are implemented in TensorFlow 2.14 and trained with Adam [24] (learning rate 2×10^{-4}, batch size 8) on one NVIDIA RTX 3070 (8 GB), which also runs all inference. Early stopping monitors the validation diffusion loss (patience 10), and the best checkpoint is tested. Seeds are fixed to 42 and the ensemble size is $M = 30$.

Metrics. For an observation y and ensemble $\{\hat{y}^{(m)}\}_{m=1}^{M}$, the primary metric is the empirical CRPS [26],

$$\mathrm{CRPS} = \frac{1}{M}\sum_{m} \left|\hat{y}^{(m)} - y\right| - \frac{1}{2M^2}\sum_{m,m'} \left|\hat{y}^{(m)} - \hat{y}^{(m')}\right|, \quad (3)$$

averaged over windows and lead times; for $M = 1$ it equals the mean absolute error (MAE). Skill is $1 - \mathrm{CRPS}/\mathrm{CRPS}_{\mathrm{SPM}}$. The central 90% interval (5th–95th ensemble percentiles) is scored by the Winkler score (WS_{90}) [7], coverage and width. Point accuracy of the ensemble median and mean is given by MAE, RMSE, mean bias error (MBE) and R^2. Generated frames are compared with observed ones by SSIM and the cosine similarity of VGG-16 features [23] (VGG-CS), as is standard in video prediction [27], with last-frame persistence as reference.

V. Results

Probabilistic accuracy. Table II shows that the joint DDIM ensemble has the lowest CRPS, 1.222 kW, 7.5% better than smart persistence and 18.4% better than the two-stage pipeline. The latter is 13.4% worse than persistence, because errors in its predicted video propagate to a PV regressor trained on true frames; the joint model denoises both outputs together and avoids this mismatch. The 18.4% gain matches the 18% reported for PV-MM-Diffusion [1] despite a 4× shorter sampler, a lighter backbone and all-sky test days, though with a different decoupled baseline. Our lower absolute CRPS mainly reflects the easier all-sky test set. Withholding the sky images raises the CRPS by 70% to 2.072 kW and doubles the Winkler score, so the images carry most of the predictive information.

TABLE II
Probabilistic Accuracy on the SKIPP'D Test Set (1,340 Windows, 2–16 min). Skill vs. Smart Persistence; Nominal Coverage 90%.

Method	CRPS (kW)	Skill	WS_{90} (kW)	$\mathrm{Cov}_{.90}$
Smart persistence†	1.321	0.0%	–	–
Two-stage pipeline†	1.498	−13.4%	–	–
Joint DDIM, PV-only input	2.072	−56.8%	20.61	–
Joint DDIM, SI+PV input	**1.222**	**+7.5%**	**10.50**	77.9%

†Deterministic: CRPS reduces to MAE; interval scores are undefined.

Calibration. The 90% intervals cover 77.9% of observations, so the ensemble is under-dispersed. Fig. 2(b,c) shows that coverage falls from 81.6% at +2 min to 75.1% at +16 min while the mean width grows from 3.69 to 6.10 kW: the spread increases with lead time, but not as fast as the errors. With $\eta = 0$ the only source of diversity is the initial noise, and 15 steps pass quickly through the high-noise regime, which is a plausible cause.

Point accuracy. As a point forecast the picture reverses (Table III). The DDIM median has the highest MAE (1.684 vs.

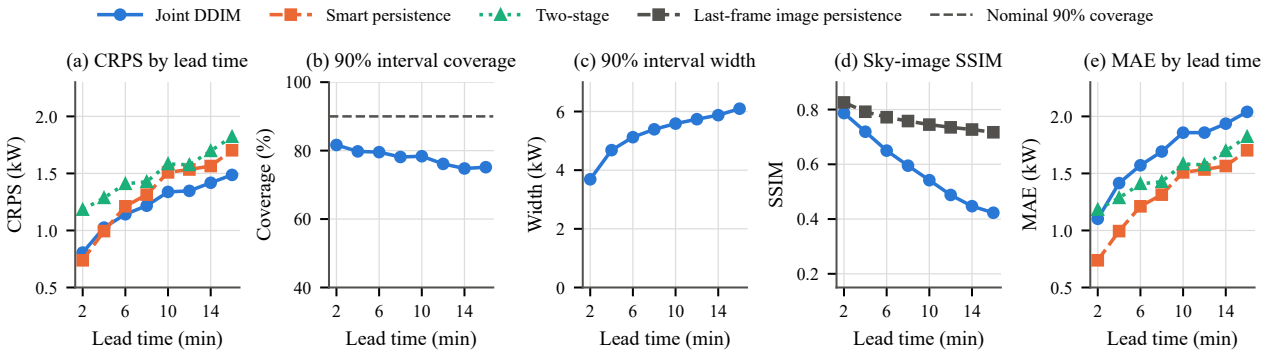


Fig. 2. Performance by lead time on the 1,340 test windows. (a) CRPS of the DDIM ensemble, smart persistence and the two-stage pipeline (equal to MAE for the deterministic baselines). (b) Coverage and (c) mean width of the DDIM 90% interval. (d) SSIM of DDIM-generated frames (first member) and last-frame persistence against the observed future frames. (e) MAE of the DDIM ensemble median, smart persistence and the two-stage pipeline.

TABLE III
POINT-FORECAST ACCURACY AND CRPS (ALL 1,340 WINDOWS; DAYTIME MAE IN BRACKETS)

Method	CRPS* (kW)	MAE (kW)	RMSE (kW)	MBE (kW)	R^2
Smart persistence	1.321	**1.321** [**1.481**]	3.087	**−0.035**	0.824
Two-stage	1.498	1.498 [1.608]	**2.838**	0.172	**0.851**
DDIM median	**1.222**	1.684 [1.899]	2.960	−0.763	0.838
DDIM mean		1.823 [2.046]	2.956	−0.837	0.839

*Of the full 30-member DDIM ensemble; equal to MAE for the deterministic baselines.

1.321 kW for persistence) but a lower RMSE than persistence and a higher R^2, i.e., many moderate errors but fewer large ones: its 99th-percentile absolute error is 12.68 kW against 14.72 kW for persistence, which fails at ramps that the image-conditioned model partly anticipates. The DDIM forecasts are biased low (MBE −0.76 kW for the median), which is most visible as afternoon dips on clear days (Fig. 3(a)–(c)); on the most variable day (f), the 90% band widens during the midday ramps while persistence reproduces each ramp one lead time late. Persistence is best at +2 min, and the relative MAE gap narrows from 49% at +2 min to 20% at +16 min (Fig. 2(e)). The CRPS rewards an ensemble whose spread covers the observation even when its centre is off, so the model's advantage lies in its distribution, not its central estimate.

Generated images and cost. The generated frames reach a VGG-CS of 0.862. They preserve the sun position and large-scale cloud layout for the first minutes and then blur and lose fine structure, and on the clear-sky example they gradually add spurious haze (Fig. 4); SSIM falls from 0.787 at +2 min to 0.423 at +16 min, whereas last-frame persistence falls only from 0.826 to 0.717 (Fig. 2(d)). At this capacity the image branch does not model cloud advection faithfully beyond a few minutes. Sampling takes 155.5 ms per window per member on the RTX 3070 (4.7 s for 30 members) and grows almost linearly with the step count (56 and 249 ms at 10 and 60 steps). The reference needs 45.39 s for 10 members on an RTX 3090 [1], about 29 times our per-member cost, owing to both sampler and network size.

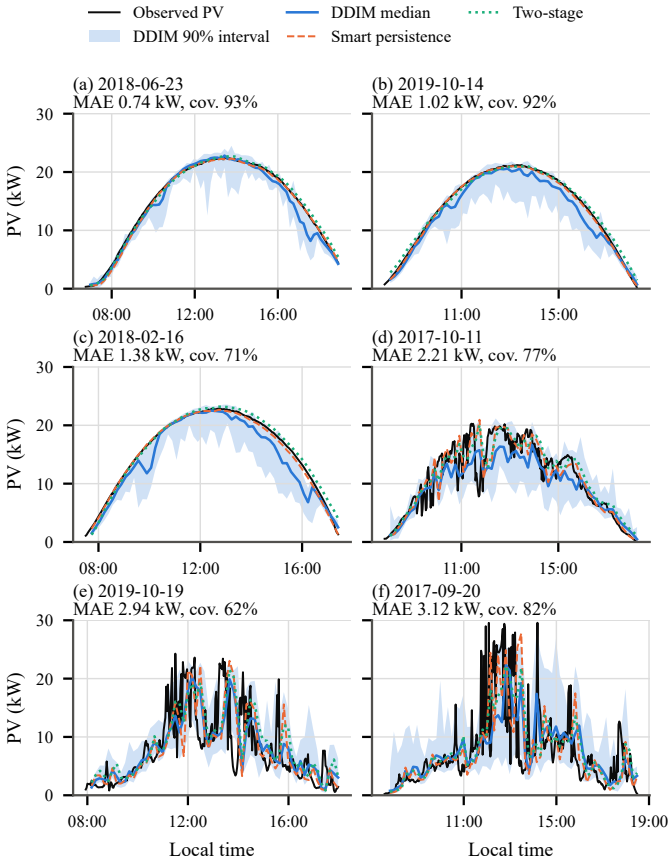


Fig. 3. 16-minute-ahead forecasts on six test days at evenly spaced ranks of PV variability, from (a) least to (f) most variable. Band: DDIM 90% interval (30 members); panel titles give the day's DDIM median MAE and coverage.

VI. DISCUSSION AND LIMITATIONS

Joint diffusion gives the best probabilistic score and clearly beats a decoupled video-then-PV pipeline, supporting the central claim of [1] on all-sky days with a 15-step sampler on one GPU. It is, however, under-dispersed, biased low and

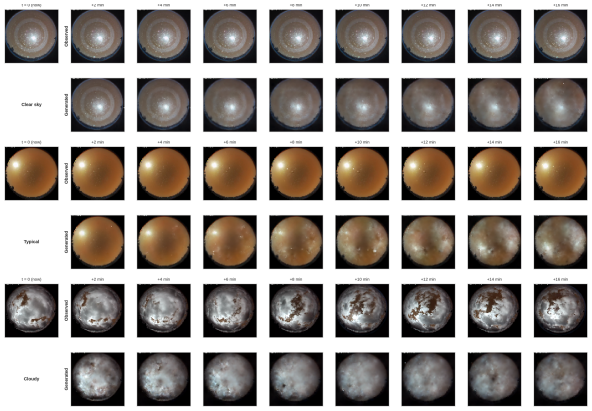


Fig. 4. Observed (upper row of each pair) and DDIM-generated (lower row, first member) sky images for the daytime test windows with the lowest, median and highest PV variability. The first column is the last observed frame.

weaker than persistence as a point forecaster. Remedies include stochastic sampling ($\eta > 0$) and larger ensembles, post-hoc recalibration, predicting the residual to smart persistence, and larger backbones or latent diffusion [19]. Whether the weaknesses stem from our reductions cannot be decided here; testing on the SkyGPT partition, sampling one model with both solvers and restoring patch-level cross-attention would separate the factors. The study covers one site, one horizon, 20 test days and one training run.

VII. Conclusion

We evaluated a compact single-GPU re-implementation of PV-MM-Diffusion sampled with 15-step DDIM. On 1,340 all-sky SKIPP'D test windows, the 30-member ensemble achieves a CRPS of $1.222\,\mathrm{kW}$, improving on smart persistence by 7.5% and on a two-stage pipeline by 18.4%, at $155\,\mathrm{ms}$ per member, but it is under-dispersed (77.9% coverage of 90% intervals), biased low and weaker than persistence as a point forecaster. Closing these calibration and bias gaps is the clearest route to operational use.

References

[1] J. Huang, B. Shao, Y. Ke, Y. Gao, and S. Mazzoni, "PV-MM-diffusion: An end-to-end multi-modal diffusion model for ultra-short-term probabilistic photovoltaic forecasting," *Applied Energy*, vol. 413, p. 127816, 2026.

[2] Y. Nie, X. Li, A. Scott, Y. Sun, V. Venugopal, and A. Brandt, "SKIPP'D: A SKy Images and Photovoltaic Power Generation Dataset for short-term solar forecasting," *Solar Energy*, vol. 255, pp. 171–179, 2023.

[3] H. Fan, W. Liu, Y. Lu, D. Liu, L. Cheng, and W. Wei, "PARA-PV: Physics-aware retrieval-augmented PV prediction based on frozen foundation model and distribution shift correction," *arXiv preprint arXiv:2607.08079*, 2026.

[4] H. Jiang, Z. Yang, Y. Li, R. Wang, J. Shi, and Y. Zhao, "A novel multimodal framework for ultra-short-term photovoltaic power forecasting via fusion of sky images and time-series data," *International Journal of Electrical Power & Energy Systems*, vol. 180, p. 112064, 2026.

[5] G. Ruan *et al.*, "SkyEvo: Generative probabilistic solar forecasting with sky image evolution," *Solar Energy*, vol. 317, p. 114996, 2026.

[6] J. Antonanzas, N. Osorio, R. Escobar, R. Urraca, F. J. Martinez-de Pison, and F. Antonanzas-Torres, "Review of photovoltaic power forecasting," *Solar Energy*, vol. 136, pp. 78–111, 2016.

[7] D. W. van der Meer, J. Widén, and J. Munkhammar, "Review on probabilistic forecasting of photovoltaic power production and electricity consumption," *Renewable and Sustainable Energy Reviews*, vol. 81, pp. 1484–1512, 2018.

[8] Y. Sun, V. Venugopal, and A. R. Brandt, "Short-term solar power forecast with deep learning: Exploring optimal input and output configuration," *Solar Energy*, vol. 188, pp. 730–741, 2019.

[9] Y. Nie, E. Zelikman, A. Scott, Q. Paletta, and A. Brandt, "SkyGPT: Probabilistic ultra-short-term solar forecasting using synthetic sky images from physics-constrained VideoGPT," *Advances in Applied Energy*, vol. 14, p. 100172, 2024.

[10] Q. Paletta, G. Arbod, and J. Lasenby, "Benchmarking of deep learning irradiance forecasting models from sky images – An in-depth analysis," *Solar Energy*, vol. 224, pp. 855–867, 2021.

[11] Q. Paletta *et al.*, "Advances in solar forecasting: Computer vision with deep learning," *Advances in Applied Energy*, vol. 11, p. 100150, 2023.

[12] W. F. Holmgren, C. W. Hansen, and M. A. Mikofski, "pvlib python: A python package for modeling solar energy systems," *Journal of Open Source Software*, vol. 3, no. 29, p. 884, 2018.

[13] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 6840–6851.

[14] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *International Conference on Learning Representations (ICLR)*, 2021.

[15] J. Ho, T. Salimans, A. Gritsenko, W. Chan, M. Norouzi, and D. J. Fleet, "Video diffusion models," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022.

[16] L. Ruan *et al.*, "MM-Diffusion: Learning multi-modal diffusion models for joint audio and video generation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 10 219–10 228.

[17] K. Rasul, C. Seward, I. Schuster, and R. Vollgraf, "Autoregressive denoising diffusion models for multivariate probabilistic time series forecasting," in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021, pp. 8857–8868.

[18] Y. Tashiro, J. Song, Y. Song, and S. Ermon, "CSDI: Conditional score-based diffusion models for probabilistic time series imputation," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 24 804–24 816.

[19] Z. Gao *et al.*, "PreDiff: Precipitation nowcasting with latent diffusion models," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.

[20] J. Leinonen, U. Hamann, D. Nerini, U. Germann, and G. Franch, "Latent diffusion models for generative precipitation nowcasting with accurate uncertainty quantification," *arXiv preprint arXiv:2304.12891*, 2023.

[21] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015, pp. 234–241.

[22] A. Vaswani *et al.*, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[23] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *International Conference on Learning Representations (ICLR)*, 2015.

[24] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *International Conference on Learning Representations (ICLR)*, 2015.

[25] C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, and J. Zhu, "DPM-Solver++: Fast solver for guided sampling of diffusion probabilistic models," *arXiv preprint arXiv:2211.01095*, 2022.

[26] P. Lauret, M. David, and P. Pinson, "Verification of solar irradiance probabilistic forecasts," *Solar Energy*, vol. 194, pp. 254–271, 2019.

[27] A. X. Lee, R. Zhang, F. Ebert, P. Abbeel, C. Finn, and S. Levine, "Stochastic adversarial video prediction," *arXiv preprint arXiv:1804.01523*, 2018.